

Taxi Tap

Digital Wallet for Minibus Taxi Fares

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PROBLEM DEFINITION AND LITERATURE REVIEW

Taxi Tap: Digital Wallet for Minibus Taxi Fares

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Contents of this submission

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1	Chapter 1: Introduction and Problem Definition	Chapter 1 (inserted from group draft)
2	Chapter 2: Literature Review and Research	Chapter 2 and Literature Review Matrix
3	Reference list (Harvard, CTU style)	List of Sources
4	Project plan (version 1)	Appendix A, with live Linear project
5	GitHub repository	Link on this page

DECLARATION OF AUTHENTICITY

We declare that this submission is our own work. Where we have used the work or ideas of others, these have been acknowledged with in-text citations and listed in the List of Sources. Any use of AI tools in preparing this work has been declared to the supervisor in line with the CTU Referencing and Style Guide (2025).

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1. INTRODUCTION AND PROBLEM DEFINITION

1.1 Background and Real World Context

Minibus taxis are very important when it comes to public transport in South Africa, providing transport for a large population of our country's work force, students, travelers and any other destination you can think of.

Passengers and Taxi Operators need a convenient and reliable way to manage and pay their taxi fares, traditional cash transactions can create challenges, including the need for passengers to carry money on their person.

There are problems getting the correct change, faked bills and difficulty handling and tracking fare payments.

As technology develops there needs to be a more efficient way to record and take transactions which would void the current issues. A digital wallet would allow users to pay through their phone or near field communication (NFC), or even things such as scanning a QR code.

The proposed project is, Taxi Tap, a digital wallet for minibus taxi fares. We aim to explore a better way to pay for taxi fares without the struggles people go through each day.

The project will focus on developing and testing a functional software prototype within the project timeline. We will use simulated funds and will not be connected to real banks, but the project will explain how collaborating with a real bank would look like.

1.2 Problem Statement

Traditional cash based taxi fares depend on passengers and minibus operators to transact fares using physical money. This creates problems when passengers don't have the correct amount of change, as well as for the minibus taxi operators. This cash method also is bad to keep track of every transaction, and may lead to incorrect money handling, theft and bad record holding.

This project aims to take the opportunity to investigate whether a digital wallet would be better for both passengers and minibus operators, as an alternative method to process and record transactions without the need of physical cash.

The challenge addressed by this project is the design and development of a functional prototype that shows how passengers could pay minibus taxi fares digitally and how taxi operators could verify those payments using a simulated online bank and QR reader.

The project will investigate how QR codes and a simulated NFC payments can be incorporated into a digital wallet application while taking record of accurate simulated balances and transaction records. The prototype will also demonstrate how taxi fares can be applied to payments.

1.3 Research Questions

The project will be guided by the following research questions:

How can a digital wallet application be designed to support minibus taxi fare payments?

How can customer and merchant modes be used to facilitate the payment and verification process?

How can QR codes and a simulated NFC payment be incorporated into the prototype?

How can a simulated boarding reader and fare recorder be used to calculate and verify taxi fare payments?

How can simulated wallet balances and transaction records be managed accurately and consistently?

How usable and reliable is the proposed system when evaluated through functional testing and user feedback?

1.4 Project Aim

The aim of this project is to design, develop and test a functional prototype of Taxi Tap, a digital wallet application intended to demonstrate an alternative method of managing and processing minibus taxi payments using QR codes, a simulated NFC payment flow and a simulated boarding recorder.

The prototype will use simulated funds and predefined fares to demonstrate the payment process for both passengers and taxi operators. The project will evaluate the system's functionality, usability and reliability while identifying the limitations and requirements that would need to be addressed before a real-world implementation could be considered.

1.5 Project Objectives

The project will pursue the following objectives:

1. Investigate the problem: Research the existing minibus taxi fare payment process and identify potential challenges and user needs.
2. Gather requirements: Identify and document the functional and non functional requirements for the customer wallet, payment process and boarding payment recorder.
3. Design the application: Develop the system, user interface designs, data structures and payment methods for the prototype.
4. Develop customer mode: Implement functionality that allows passengers to view their wallet balance and pay for taxi fares.
5. Develop merchant mode: Implement functionality that allows taxi operators to view payment requests and verify successful transactions.
6. Implement the payment flow: Develop a QR code payment process and a simulated NFC payment method suitable for demonstrating the proposed system.
7. Develop the simulated boarding recorder: Create a software application boarding reader that applies a fare table and supports payment verification.
8. Manage simulated transactions: Ensure that wallet balances are updated correctly following successful payments and that transaction outcomes can be recorded and reviewed.
9. Test and evaluate the prototype: Conduct functional testing and usability throughout development to identify errors and assess whether the system meets its requirements.
10. Document limitations and future improvements: Explain the limitations of the prototype, including the process to use real funds, bank integration and physical payment hardware, and identify considerations for future development.

1.6 Project Scope and Limitations

1.6.1 Project Scope

1. A Customer wallet: A customer interface for viewing a simulated balance and doing fare payments.
2. A Merchant mode: A merchant interface for receiving and verifying simulated fare payments.
3. QR payment: A QR process for completing a simulated payment.
4. Simulated NFC payment: A software demonstration of the intended NFC payment process without requiring physical NFC payment hardware.
5. Simulated boarding recorder: A software based reader that demonstrates how a boarding payment could be made and verified.
6. Fare table: A predefined table of taxi fares used to determine the amount payable for a selected route or fare category.
7. Simulated funds: A controlled test balance in our application used to demonstrate payments without transferring real world money.
8. Transaction management: Recording transaction details and displaying payment outcomes.
9. Testing: Testing the functionality and usability of the prototype and documenting the results.

1.6.2 Project Limitations

1. No real financial transactions: The wallet will use simulated funds. It will not transfer, store or settle actual customer money in real life.
2. No physical payment hardware: The boarding reader and NFC functionality will be demonstrated through software rather than physical devices.
3. Limited fare data: The system will use a predefined fare table for demonstration and testing. It will not automatically obtain live fare information from real taxi associations, we will be using our own dummy data.
4. Limited development timeframe: The project must be completed within the four week period. The group will prioritise the wallet, payment, boarding reader and testing rather than attempting to implement every feature that a commercial product might require.
5. Regulatory limitations: A real life payment service would require an assessment of applicable South African financial and payment regulations and appropriate permissions with an authorised or licensed payment partner. The prototype itself will not provide a live regulated payment method like Capitec for example.

1.7 Significance of the Project

The Taxi Tap project is significant because it explores the potential application of digital payment technology to a much used way of public transportation in South Africa.

For passengers, the proposed system could demonstrate an alternative to cash payments and provide a way to view payment outcomes and transaction records.

For taxi operators, the prototype could demonstrate how digital payments may be verified and recorded, potentially reducing reliance on physical cash handling and making payment records easier to review and to record income.

From a software engineering perspective, the project provides an opportunity to apply practical development skills, interface design, application development, transaction processing, data management, testing, documentation skills etc.

We Will be using simulated funds instead of real funds from real banks, the group can test payment workflows without handling real money or any real risk.

This project does not claim that digital payments will be suitable for every passenger or taxi operator...

1.8 Target Stakeholders

Passengers :

Passengers are the primary users of the wallet. They would use the app to view their balance, fare payments and confirm whether transactions were successful.

Their main interests include ease of use, clear payment confirmation, payment amount calculations and confidence that payments are processed correctly without needing to use physical money.

Operators and Drivers :

Taxi operators and drivers would use the merchant mode and simulated boarding recorder. They would use the app to initiate or verify fare payments and confirm that a passenger's transaction has been completed successfully.

Their main interests include a quick boarding process, reliable payment verification and better income recordings

Taxi Associations :

Taxi associations are a big stakeholder because they represent taxi operators and may provide information about routes, fare structures and operating practices.

Their involvement would help the group understand real requirements.

1.9 User Stories

1. Passenger Wallet: As a passenger, I want to view my wallet balance so that I know how much money is available to pay for taxi rides.
2. Digital Payments: As a passenger, I want to pay for my taxi fare using a QR code or NFC payment so that I can complete a fare transaction digitally.
3. Fare Calculation: As a passenger, I want the application to show me the correct taxi fare based on a fare table so that I know how much I need to pay for my journey in total or before I even get on the taxi based on my current location.
4. Payment Verification: As a taxi operator, I want to use the boarding recorder to verify a passenger's payment so that I can confirm whether the fare has been paid before allowing the passenger to board.
5. Transaction History: As a passenger, I want to view my transaction history so that I can keep track of my previous taxi fare payments and their payment statuses.

2. LITERATURE REVIEW AND RESEARCH

2.1 Introduction

Cash remains the default way to pay for a minibus taxi trip in South Africa, even as taxi associations, banks and fintech start-ups pilot card, app and QR alternatives. Local and Kenyan experience shows that the technology is rarely the main obstacle. Adoption depends on whether owners, drivers, associations and commuters each gain something from the change, and on whether the system fits how taxis actually operate: passengers board at ranks or roadside stops, alight anywhere and pay by zone or route. Sections 2.2 and 2.3 set out why the industry still runs on cash and what earlier cashless fare projects learned. Sections 2.4 and 2.5 evaluate seven current and past solutions, section 2.6 covers the technologies that suit a phone-based prototype, and section 2.7 states the gap Taxi Tap addresses.

2.2 The minibus taxi industry and cash

Woolf and Joubert (2013:284-293) describe South African paratransit as an informal but well-organised system that grew out of apartheid-era restrictions and later deregulation, and that now dominates the local public transport market. They argue that the industry should be understood as many individual operators and associations trying to make a living, rather than a single problem to be replaced. This matters for this project because any payment system must work for owners, drivers and associations, not only for commuters.

The industry is large. In 2016 the industry was estimated at about 200 000 taxis carrying around 15 million passengers a day and generating more than R90 billion a year in fares (*Mail & Guardian* 2016). Most of this revenue is still collected as cash.

Schalekamp, McLaren and Behrens (2017) explain how cash shapes behaviour in the industry. Drivers typically pay the owner a fixed amount and keep the rest, which encourages reckless driving to maximise trips, while owners have little reliable revenue data and struggle to fund fleet renewal. Their multi-criteria evaluation of cashless options found that the weakest options either required passengers to hold a bank account or relied on a stand-alone fare system. Two lessons follow for Taxi Tap: the wallet must not depend on a bank account, and it must not lock commuters into a single-purpose card.

2.3 Cashless fare collection in African paratransit

Tinka and Behrens (2019) reviewed 24 cashless fare collection initiatives across Sub-Saharan Africa and found that most minibus initiatives were unsuccessful, while motorcycle-taxi initiatives did better. Aruho, Behrens, Mitullah and Kamau (2021) studied matatu cashless projects in Nairobi through stakeholder interviews and a crew survey. They found no single initiator, and identified four implementation problems: low knowledge of the system, weak enforcement, no interoperability between systems, and a lack of trust.

Kenya's BebaPay is the best-known failure. Launched by Google and Equity Bank in 2013, it used prepaid NFC cards read by conductors' Android phones (Joseph 2015). Commuters had to carry a card used only for transport, and crews preferred cash because cashless collection sent all fares directly to owners and removed the money crews could keep (*The Standard* 2015; Joseph 2015). The service was discontinued in 2015. Read through the Technology Acceptance Model (Davis 1989:319-340), BebaPay asked two groups to change behaviour at once without a clear benefit to either: commuters gained little usefulness

from a single-purpose card, and crews lost income. Taxi Tap therefore needs a benefit for drivers (no cash counting, no change disputes, a visible shift record) as well as for owners and commuters.⁴

2.4 Review of existing solutions

Seven solutions were reviewed: five South African taxi fare systems, one informal-trader wallet and one failed international case. Each is assessed for what it does well, where it falls short, and the lesson for this project.

2.4.1 Codeta SAPay (Western Cape)

The Cape Organisation for the Democratic Taxi Association (Codeta) launched a cashless pilot in June 2026 on routes from Khayelitsha and Mfuleni to Killarney, where commuters pay with cards scanned on entry (*Western Cape Government* 2026; George 2026). Reporting on the SAPay system describes free reloadable smart cards, an app with self-scan QR codes, bank card acceptance, an offline mode for the driver's phone, and in-vehicle cameras (*The Mercury* 2026).

Strengths: association-led, so adoption is driven from inside the industry; several payment methods; an offline mode; provincial government support.

Limitations: limited to one association's routes; the coverage focuses on fares and tax compliance rather than a wider wallet; cameras and cards add operating cost.

Lesson for Taxi Tap: association ownership matters for adoption, and a QR self-scan option works alongside cards.

2.4.2 Santaco Wealth on Wheels (Eastern Cape)

WOW is led by the Eastern Cape Transport Tertiary Cooperative, Santaco's business arm, with Ecentric Payment Systems and FORUS Digital. It plans to fit about 1 000 taxis with GPS trackers, cameras, Wi-Fi and payment devices, and includes point-of-sale for informal merchants (*Pretoria News* 2024). In late 2025 Santaco registered WOW Mobile, an industry-owned mobile network, and earmarked March 2026 for the cashless pilot (*Daily Dispatch* 2025).

Strengths: strong industry backing; an integrated ecosystem including connectivity and merchants; a national model ambition.

Limitations: a hardware-heavy rollout per vehicle; long timelines (announced 2024, pilot set for 2026); the cooperative itself argues that formalisation must come before cashless payments.

Lesson for Taxi Tap: a merchant and connectivity layer is valuable, but the per-vehicle hardware cost is a barrier, which supports using phones as readers.

2.4.3 Waxd and Botransa (North West)

Waxd runs an automatic fare collection system with the Botransa Taxi Association in Swartruggens, with phase 1 running since November 2024 (*TechFinancials* 2025). The company reports that drivers no longer count cash while driving or argue about change, and that owners collect fares without handling cash.

Strengths: a working rural deployment; designed with drivers, owners and associations; measurable benefits reported by drivers.

Limitations: a small, single-town pilot; a proprietary point-of-sale ecosystem; little public evidence beyond company statements.

Lesson for Taxi Tap: co-designing with all stakeholder groups is what Waxd credits for adoption, and Taxi Tap's requirements gathering should follow this.

2.4.4 Loop (Cape Town)

Loop is a Cape Town start-up that lets commuters scan a QR code displayed in the taxi and pay through WhatsApp (Chat-to-Pay), with Visa Tap to Phone added in 2023 so drivers accept card taps on their own smartphones (Melapi 2023; *NFCW* 2023). The pilot started with about 70 taxis on the Mitchells Plain to Bellville route.

Strengths: no dedicated hardware; uses familiar channels (WhatsApp, bank cards); the driver's phone becomes the terminal.

Limitations: commuters need a smartphone, data and a bank card or account; per-route onboarding.

Lesson for Taxi Tap: the driver's phone as reader is proven in South African taxis, which supports Taxi Tap's phone-based reader design.

2.4.5 Tap-I-Fare and FairPay (earlier card systems)

In 2012 the Peninsula Taxi Association piloted Tap-I-Fare, an EMV contactless prepaid card backed by Absa, in 42 taxis on the Cape Town city to V&A Waterfront route (*Brand South Africa* 2012). In 2016 Santaco's trading arm introduced the FairPay FairCard, reporting earlier violence over rival card systems (*Mail & Guardian* 2016).

Strengths: early proof that card payment can work in a taxi; fare discounts as an incentive.

Limitations: card-only; issuer fees; neither became a national standard; competing systems caused conflict.

Lesson for Taxi Tap: interoperability and industry buy-in matter more than the card technology itself.

2.4.6 Street Wallet (informal traders)

Street Wallet gives street vendors, car guards and informal traders a QR lanyard card linked to their profile, so customers can pay digitally without the trader owning a card machine. It raised \$350 000 in 2025 and charges about 5% per street-market transaction (*TechCabal* 2025).

Strengths: very low-cost acceptance (a printed QR card); designed for people without card machines or reliable data.

Limitations: serves traders only, not transport; the payer still needs a payment app; a relatively high fee.

Lesson for Taxi Tap: a printed QR is enough for a trader to accept payments, which supports Taxi Tap's merchant mode.

2.4.7 BebaPay (Kenya, discontinued)

BebaPay is reviewed in section 2.3. It is included here because it shows how a technically sound NFC system can still fail.

Strengths: strong technology partners; free reader software on Android phones.

Limitations: a single-purpose card; no benefit for crews; discontinued after about two years (Joseph 2015).

Lesson for Taxi Tap: design for driver incentives and for a wallet that is useful beyond one ride.

2.5 Comparison of existing solutions

Table 1 summarises the reviewed solutions. Table 2 compares them against the features that the problem in Chapter 1 requires.

Table 1: Existing cashless solutions reviewed

Solution	Region and status	Payment method	Reader / acceptance	Bank account needed
Codeta SAPay	Western Cape, pilot June 2026	Smart card, app QR, bank card	Device in taxi, cameras	No (card)
Santaco WOW	Eastern Cape, pilot planned 2026	WOW card or wallet	In-vehicle devices, GPS, Wi-Fi	No (card/wallet)
Waxd + Botransa	North West, since Nov 2024	Waxd cashless system	Waxd point of sale	Not stated
Loop	Cape Town, since 2022–23	WhatsApp QR, card tap	Driver's smartphone	Yes (card/bank)
Tap-I-Fare / FairPay	Cape Town 2012 / Gauteng 2016	Prepaid card	Handheld or fixed reader	No (prepaid)
Street Wallet	SA, traders, 2021–	QR lanyard	Printed QR, payer's app	Payer: yes
BebaPay	Nairobi, 2013–2015 (closed)	Prepaid NFC card	Conductor's Android phone	No

Table 2: Comparison of existing solutions against Taxi Tap (? = not publicly stated; Design = documented but not built in this project)

Feature	Codeta	WOW	Waxd	Loop	Street Wallet	Taxi Tap
Fares set at boarding for zone and route fares	Partial	Partial	Partial	No	No	Yes
One wallet for taxis and traders	No	Partial	No	No	No	Yes
Feature	Codeta	WOW	Waxd	Loop	Street Wallet	Taxi Tap
Works without a bank account	Yes	Yes	?	No	No	Yes
No dedicated in-vehicle hardware	No	No	No	Yes	Yes	Yes
Owner sees takings per taxi and driver	Yes	Yes	Yes	Partial	No	Yes
Phone-free fallback for commuters	Yes	Yes	?	No	No	Design
Data-light / zero-rated use	Partial	Partial	?	No	No	Design

The comparison shows that current South African projects are converging on association-led, card-based systems with hardware in each vehicle (Codeta, WOW, Waxd). These carry real per-vehicle costs and are tied to one association. Loop avoids hardware by using the driver's phone, but excludes commuters without a bank card or data. No reviewed system offers a single wallet that both taxi operators and informal traders can accept with only a phone and a printed QR.

2.6 Related technologies

QR codes. QR payment needs only a camera and a printed or on-screen code, which is why Loop and Street Wallet use it. It is the most widely supported option for a web app and will be Taxi Tap's main payment path.

NFC and Web NFC. The W3C Web NFC Community Group specification lets web pages read and write NFC tags through the NDEFReader interface, only in secure (HTTPS) contexts and with user permission (*W3C Web NFC Community Group* n.d.). Chromium documents the implementation as Android-only (*Chromium Projects* n.d.), so iOS users cannot use it. Taxi Tap will therefore treat NFC tag reading on Android as an optional feature with a simulated-tap fallback, and keep payment cards and wristbands as design-only, as agreed in the topic acceptance.

Tap to Phone (softPOS). Visa Tap to Phone lets a driver's smartphone accept contactless cards (*NFCW* 2023). It requires a card network partner and certification, so it is outside the scope of a student prototype but supports the case for phone-based readers.

PayShap. The South African Reserve Bank and BankservAfrica launched PayShap in March 2023, a low-value real-time payment system that uses a phone-number proxy (ShapID) (*BusinessTech* 2023). It is a likely future route for bank-neutral top-ups and is recorded as a design-only integration.

Progressive web apps (PWA). A PWA runs in the phone's browser and can be installed to the home screen without an app store. This suits the project's four-week timeline and lets the team test on both Android and iOS from one codebase.

Boarding time. A Visa study cited by Mondato (n.d.) found digital bus payments took about 2.6 seconds against 4.2 seconds for cash. This gives a reference point for Taxi Tap's non-functional requirement on tap-to-confirm time.

2.7 Research gap

The literature and the industry review point to the same gap. Existing systems either need dedicated hardware in every taxi (Codeta, WOW, Waxd), need a bank card and data (Loop), or serve only traders (Street Wallet). Earlier projects failed when they ignored drivers' incentives, lacked interoperability, or tied commuters to a single-purpose card (Tinka & Behrens 2019; Aruho et al 2021; Joseph 2015). None of the reviewed systems sets zone and route fares at boarding for passengers who board and alight anywhere, using only phones and printed codes, inside a wallet that informal traders can also accept.

Taxi Tap addresses this gap with a bank-neutral wallet, a phone-based boarding reader with route fare tables, and a merchant mode with a printed QR. The gap links to the requirements as follows:

Gap identified	Evidence	Requirement (draft IDs)
Fares must be set at boarding without tap-off	Schalekamp et al (2017); taxi review	FR-06, FR-07, FR-11
No dedicated in-vehicle hardware	WOW and Codeta hardware costs; Loop	FR-07 (phone as reader)
Must not depend on a bank account	Schalekamp et al (2017); Loop limits	FR-03 (platform wallet)
Drivers need a benefit	BebaPay; Aruho et al (2021)	FR-08, FR-10
Owners need reliable takings data	Schalekamp et al (2017); Waxd	FR-14
One wallet across taxis and traders	Street Wallet; Table 2	FR-09
Trust and reliability	Aruho et al (2021)	FR-04, NFR-05

2.8 Summary

Research shows that cash in minibus taxis persists for structural reasons, and that cashless projects fail when they ignore drivers, interoperability and trust. Current South African projects are association-led and hardware-heavy, while phone-based options exclude commuters without bank cards. These findings shape Taxi Tap's requirements in Chapter 3: a bank-neutral wallet, a phone-based reader with fare tables set at boarding, benefits for drivers, and a merchant mode for traders, with hardware, bank linking and zero-rated data kept as design-only.

2.9 Literature review matrix

Table 3: Literature review matrix

Source	Type	Focus	Key finding	Use in project
Woolf & Joubert (2013)	Journal (Cities)	SA paratransit	Informal but organised industry of individual operators	Stakeholder framing
Schalekamp et al (2017)	Conference (SATC)	Cashless options in SA	Bank-account and stand-alone systems scored lowest	Bank-neutral wallet
Tinka & Behrens (2019)	Conference (SATC)	24 African CFC cases	Most minibus initiatives failed	Risk and design lessons
Aruho et al (2021)	Conference (SATC)	Nairobi matatus	Knowledge, enforcement, interoperability, trust	NFRs and adoption
Davis (1989)	Journal (MIS Quarterly)	Technology acceptance	Usefulness and ease of use drive adoption	Evaluation lens
Joseph (2015); The Standard (2015)	Industry / news	BebaPay	Crews resisted; single-purpose card	Driver incentives
Western Cape Govt (2026); The Mercury (2026)	Government / news	Codeta SAPay	Association-led cards, QR, offline mode	Competitor review
Pretoria News (2024); Daily Dispatch (2025)	News	Santaco WOW	Hardware-heavy, includes merchants and MVNO	Competitor review
TechFinancials (2025)	News	Waxd + Botransa	Co-design with all stakeholders	Requirements approach
Melapi (2023); NFCW (2023)	News	Loop	Driver phone as terminal; WhatsApp QR	Reader design
TechCabal (2025)	News	Street Wallet	Printed QR for traders	Merchant mode
W3C (n.d.); Chromium (n.d.)	Specification	Web NFC	HTTPS only; Android only	NFC scope decision
BusinessTech (2023)	News	PayShap	Real-time low-value payments by phone number	Future bank linking

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APPENDIX A: PROJECT PLAN (VERSION 1)

The live plan is kept in Linear (epics, stories and issues with a timeline view, sprints and owners) and is reviewed at every sprint boundary. This appendix is the baseline as at 9 October 2026.

A.1 Scope baseline

As set by the topic acceptance conditions, the group will build the wallet app (customer and merchant modes), the QR payment flow with optional NFC tag reading, and a simulated boarding reader with route fare tables, using simulated funds only. Reader hardware, cards and wristbands, bank linking and zero-rated data are design-only, and a licensed payment partner would hold real money in production. Testing starts in Sprint 1.

A.2 Team roles

Member	Role	Builds (Git evidence)	Documents
Lethukuthula Gumede (LG)	Group Lead, UI/UX designer, DevOps, backend support	UI design and wireframes; repo, CI, HTTPS deployment, releases; Sprint 1 backend (auth, organisations, wallet, ledger) while Londa is unavailable	Chapters 2, 3 (with CD) and 4; project plan, registers, progress reports
Londa Mavimbela (LM)	Backend developer	Payment API, QR tokens, fare engine, merchant payments, seed data; reviews Sprint 1 backend	Originally Chapters 2 and 3; covered by LG and CD for now
Christoffel Du Toit (CD)	Frontend developer, QA, final compiler	Customer, merchant and boarding reader screens, QR scanner, NFC (Could)	Chapters 1, 3 (with LG) and 5; test workbooks; compiles each submission

Londa's documentation is covered by Lethukuthula and Christoffel while Londa is unavailable; Londa remains the backend owner. Every member still commits code in every sprint.

Table 4: Team roles

A.3 Weekly milestones

Milestone	Due	Deliverables
Week 1 – Proposal	2 Oct (done)	One-page proposal; topic accepted with minor changes
Week 2 – Problem and literature	9 Oct, 12pm	Ch1, Ch2, references, project plan v1, GitHub repo
Week 3 – Requirements and design	16 Oct, 12pm	Ch3, SRS, Ch4 4.2–4.5, schema script and first commits from all members, revised Ch1–2, progress report
Week 4 – Build and test	23 Oct, 12pm	Working app (Must requirements), walkthrough video, Ch4 4.6–4.8, Ch5, test workbooks, full Ch1–5 draft, demo script, contribution forms
Final	29 Oct	Dissertation, source code, evidence folders, presentation and live demo; Turnitin below 15%

Table 5: Weekly milestones

A.4 Gantt chart

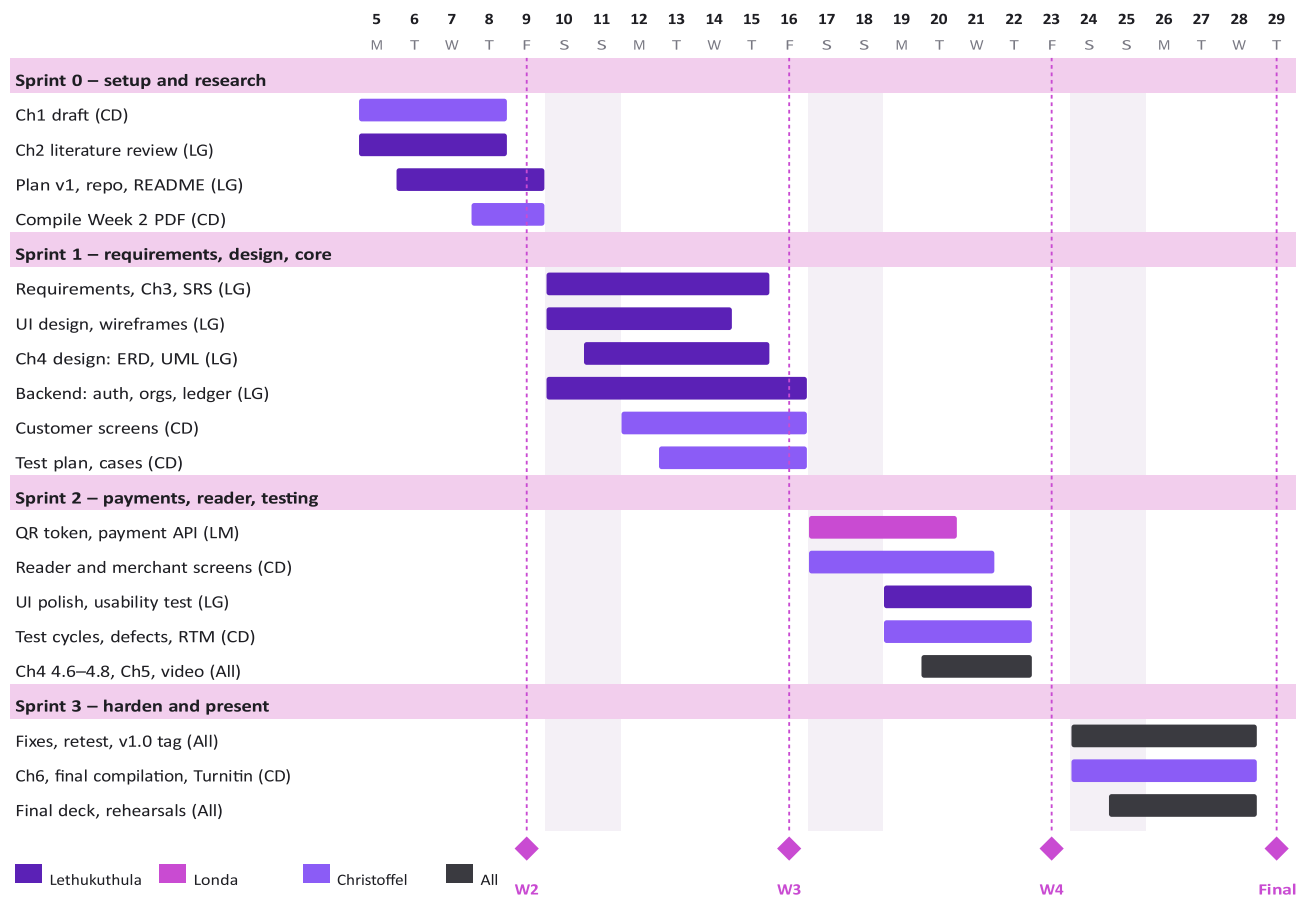


Figure 1: Project Gantt chart, 5 to 29 October 2026 (pink diamonds: submission deadlines; shaded columns: weekends). Group 4, 2026. (Project plan).

A.5 Task breakdown (epics → stories → issues)

The backlog holds 10 epics, 18 user stories and 66 issues (TT-01 to TT-66). Each issue becomes one GitHub issue, one feature branch and one pull request. The full list with owners, sprints, estimates and requirement links is kept in Linear.

Epic	Example user story	Issues	Lead	Sprints
E1 Setup and DevOps	As the team, we want one protected repository so that every member's work is tracked	TT-01–08	LG	0–3
E2 Accounts and organisations	As an association, I want owners, drivers and taxis linked to me so that payments land in the right place	TT-09–13	LG/LM	1
E3 Wallet and ledger	As a customer, I want to top up demo funds so that I can pay for rides	TT-14–18	LG/LM	1
E4 QR and NFC payments	As a driver, I want to scan a customer's code so that payment is taken in seconds	TT-19–24	LM/CD	2
E5 Boarding reader and fares	As a driver, I want the fare applied when a passenger scans so that I never handle cash	TT-25–32	CD/LM	1–2

E6 Merchant mode	As a trader, I want to request an amount and get paid without a card machine	TT-33–34	CD/LM	2
E7 Owner and driver views	As an owner, I want takings per taxi and driver so that I no longer rely on verbal reports	TT-35–37	CD	2–3
Epic	Example user story	Issues	Lead	Sprints
E8 Testing and evaluation	As the team, we want testing to run from Sprint 1 so that it does not fall in the last week	TT-38–45	CD	1–3
E9 Dissertation chapters	As the team, we want each chapter drafted a week before it is marked final	TT-46–55	LG/CD	0–3
E10 PM, UI design and presentation	As the team, we want visible planning so that the supervisor can follow progress	TT-56–66	LG	0–3

Table 6: Task breakdown by epic

A.6 Basic risk list

ID	Risk	L	I	Mitigation	Owner
R01	Scope too large for four weeks	H	H	MoSCoW; only Must items until 23 Oct; supervisor conditions define the build	LG
R02	Taxi association access delayed	H	M	Fallback: published studies, scenario analysis, permitted observation	CD
R03	Data collected from people without permission	M	H	No interviews or tests before ethics confirmation and consent	CD
R04	QR scanning unreliable on some phones	M	H	Test on Android and iOS in Sprint 1; manual code fallback	CD
R05	Web NFC not available on iOS	H	L	QR is the main path; NFC is a Could item	CD
R06	Uneven Git contributions	M	H	Each member owns coding issues; weekly check	LG
R07	Testing squeezed into the last week	M	H	Unit tests from Sprint 1; test cycles in Sprints 1 and 2	CD
R08	Turnitin similarity of 15% or more	M	H	Own words, correct citations; check by 22 Oct	CD
R09	Live demo fails	M	H	Seed and reset script, recorded walkthrough, spare phone	LG
R10	Member unavailable during a sprint	M	M	Chapters reassigned; paired knowledge per module; daily stand-up	LG
R11	Double charge or wrong balance	L	H	Atomic ledger with idempotency keys; reliability test	LM

Table 7: Basic risk list (L = likelihood, I = impact)

